23070521135 - Sharvayu Zade

```python
tf.keras.datasets.mnist.load_data()
```

[2]

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 4s 0us/step

((array([[[0, 0, 0, ..., 0, 0, 0],
          [0, 0, 0, ..., 0, 0, 0],
          [0, 0, 0, ..., 0, 0, 0],
          ...,
```

```python
import matplotlib.pyplot as plt
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, models
```

[3]

```python
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
x_train, x_test = x_train / 255.0, x_test / 255.0
x_train = x_train[..., tf.newaxis]
x_test = x_test[..., tf.newaxis]
```

[4]

```python
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])
```

[5]

```
e:\Sharvayu data\Malware\Symbiosis Nagpur SIT\7th SEM\DL LAB\DeepLearning LAB\.venv\Lib\site-packag
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```python
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

[6]

File Edit Selection View Go Run Terminal Help

p6.ipynb

E: > Sharvayu data > Malware > Symbiosis Nagpur SIT > 7th SEM > DL LAB > DeepLearning LAB > Practical 6 > p6.ipynb > M↓ 23070521135 - Sharvayu Zade

Generate + Code + Markdown | Run All Clear All Outputs | Outline ...

```
Epoch 1/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 18s 9ms/step - accuracy: 0.9555 - loss: 0.1488 - val_accuracy: 0.9824 - val_loss: 0.0561
Epoch 2/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 15s 8ms/step - accuracy: 0.9858 - loss: 0.0471 - val_accuracy: 0.9894 - val_loss: 0.0323
Epoch 3/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 19s 10ms/step - accuracy: 0.9899 - loss: 0.0331 - val_accuracy: 0.9895 - val_loss: 0.0296
Epoch 4/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 25s 13ms/step - accuracy: 0.9926 - loss: 0.0235 - val_accuracy: 0.9901 - val_loss: 0.0307
Epoch 5/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 39s 12ms/step - accuracy: 0.9941 - loss: 0.0184 - val_accuracy: 0.9925 - val_loss: 0.0239
```

```
<keras.src.callbacks.history.History at 0x273e91c7e10>
```

```python
predictions = model.predict(x_test[:5])

fig, axes = plt.subplots(1, 5, figsize=(10, 2))
for i in range(5):
    axes[i].imshow(x_test[i].squeeze(), cmap='gray')
    axes[i].set_title(f"Pred: {np.argmax(predictions[i])}")
    axes[i].axis('off')
plt.show()
```

[7]

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 136ms/step
```